

Aviyan Dhital

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EDUCATION

Texas State University

Bachelor of Science in Computer Science

Minor: Applied Mathematics | GPA: 3.80 | 59 credits

San Marcos, TX

Fall 2024 - Spring 2028

Relevant Coursework: Discrete Mathematics, Data Structures & Algorithms, Assembly Language, Probability & Statistics, Object-Oriented Programming, Computing System Fundamentals

Honors: President's List (Fall 2024, Spring 2025), Dean's List (Fall 2025)

RESEARCH EXPERIENCE

Undergraduate Research Assistant, Ingram School of Engineering

May 2026 - Present

Advisor: Dr. Sujata Mandal

Hydrothermal Carbonization of Biomass

- Built an end-to-end Python pipeline (pandas, statsmodels, scipy, scikit-learn) analyzing hydrochar formation across a two-temperature by five-residence-time reaction matrix.
- Established via two-way ANOVA that residence time drives hydrochar pH ($p=0.034$) while temperature governs C/O ratio ($p=0.036$); regression models explained 73 to 82 percent of variance in yield, pH, and C/O ratio.
- Correlated EDS elemental composition to carbonization degree (carbon $r=+0.998$) and quantified an inverse carbon-quality to mass-yield tradeoff ($r=-0.860$, $p=0.0014$).

Flood-Driven Arsenic Contamination Modeling

- Built a six-module Python pipeline pulling USGS water-quality and discharge data for site 08171350, with automated bounds validation, outlier flagging, and gap interpolation.
- Engineered 22 lag, rolling-mean, and flood-phase features for a Random Forest classifier predicting arsenic exceedance of the 10 ug/L EPA limit, evaluated by stratified cross-validation.
- Identified a two-day lag between turbidity spikes and arsenic peaks, defining an early-warning window for treatment response.

Undergraduate Research Assistant, Department of Physics

January 2026 - April 2026

- Developed Python-based linear regression models on experimental datasets and evaluated model performance using R-squared and residual analysis.
- Automated data processing and visualization workflows to replace manual spreadsheet analysis.

PROFESSIONAL EXPERIENCE

Teaching Assistant, Department of Mathematics

August 2026 - Present

- Support students in College Algebra by explaining concepts and problem-solving strategies in office hours.
- Provide individualized academic support and coordinate with the instructor on course activities.

Paper Grader, Department of Mathematics

May 2026 - July 2026

- Graded student assignments against established rubrics, maintaining consistency across submissions.
- Provided written feedback to help students strengthen conceptual understanding.

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, TypeScript, SQL

Libraries/Frameworks: Pandas, NumPy, Matplotlib, scikit-learn, SciPy, statsmodels, seaborn

Tools: Git, VS Code, Jupyter

Operating Systems: Windows, macOS, Linux

PROJECTS

Air Quality Index (AQI) Forecasting System | Python, Machine Learning

Dec 2025 - Present

- Built an AQI forecasting system on historical environmental data, reducing prediction error by 15 to 20 percent against a baseline regression model.
- Cleaned and processed 10k+ data points and evaluated multiple model classes to diagnose prediction errors.

Expense Tracker Web Application | JavaScript, Tailwind CSS, Firebase

Sept 2025 - Present

- Built a responsive expense-tracking web app with Firebase authentication (Email, Google, Phone/OTP).
- Implemented a dashboard to add, edit, delete, and split expenses, validated across 100+ test transactions.

CLUBS AND ACTIVITIES

Alpha Lambda Delta Honor Society, Member

2024 - Present

- National honor society recognizing academic excellence among first-year students.

Bobcats LEAD Emerging Leader

2025

- Applied leadership concepts through team-based discussions and workshops on communication and collaboration.